

ECON211: Economic Statistics – Midterm

Prof. Vladyslav Nora

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- Time: 50 minutes
- Instructions: Write the final answers after each question; circle the final answers clearly; provide solutions in the back of the exam or on the scratch paper.

Emergency Response Times

A city's emergency medical services department analyzes its response times. They find that the time T (in hours) it takes for the first respondents to arrive after a call follows a distribution with probability density function given below:

$$f_T(t) = \begin{cases} 2 - 2t, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

A response is officially considered "on time" if the responders arrive within first 20 minutes after a call.

1. (3 points) Find the probability that a single call is answered "on time."
2. (4 points) Let $F_T(t)$ be the CDF of the response time T . Write its functional form for $0 \leq t \leq 1$.
3. (4 points) Find the expected response time $E[T]$.
4. (5 points) Find the variance of the response time $Var(T)$.
5. (3 points) Calculate the **median** response time m . You can leave your answer in terms of a quadratic equation (do not need to solve it).

Miscellaneous problems

1. (3 points) A survey of 100 students shows that 60 like Coffee, 40 like Tea, and 20 like both. If a student is chosen at random and they like Coffee, what is the probability they also like Tea?
2. An urn has 3 white and 1 black ball. You take out 2 balls, and let X be the number of white balls that you take out.
 - (a) Suppose that you take out balls without replacement.
 - (2 points) Identify the name of the distribution of X and its parameters.
 - (2 points) What is the probability that $X = 1$?
 - (2 points) Find $E[X]$.
 - (b) Suppose that you take out balls with replacement.
 - (2 points) Identify the name of the distribution of X and its parameters.
 - (2 points) What is the probability that $X = 1$?
 - (2 points) Find $E[X]$.
3. You pick an integer from 1 to 10. A random integer then is chosen, and if it matches yours, you win \$9; else, you lose \$1 (win -1).
 - (3 points) Find the expected amount you will win.
 - (3 points) Find variance of the amount you will win.
4. (5 points) A couple has children until they have at least one boy and at least one girl. Assuming $P(\text{Boy}) = 0.5$, find the expected number of children this couple will have.
5. (6 points) Alice and Bob each independently choose 3 out of 10 different classes (without replacement). Find the expected number of classes they take together (Hint: use the decomposition into the sum of indicator random variables).